

3. A rugby player kicks a ball up into the air.



Ignore the effects of air resistance when answering the following questions.

- (a) Place **one** tick (✓) in each row to show what happens to each property as the ball **rises**. One row has been completed for you. [4]

Property of the ball	Stays constant	Gets smaller	Gets larger
acceleration	✓		
speed			
kinetic energy			
potential energy			
total energy			



- (b) (i) The rugby ball has a mass of 0.46 kg.
Use the equation:

$$\text{resultant force} = \text{mass} \times \text{acceleration}$$

to calculate the force on the ball due to gravity as it moves through the air. [2]
(acceleration due to gravity, $g = 10 \text{ m/s}^2$)

resultant force = N

- (ii) The rugby ball rises 15 m into the air.
Use your answer in (b)(i) and the equation:

$$\text{work done} = \text{force} \times \text{distance}$$

to calculate the work done by gravity on the ball. [3]
Give the correct unit.

work done =

unit =

- (iii) John says the resultant force calculated in (b)(i) is the same as the weight of the ball.
He thinks this because:

$$\text{weight} = \text{mass} \times \text{gravitational field strength}$$

Explain whether you agree with John. [1]

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